

NETWORKING FOR HACKERS

Networking 101

▲ Networking 101 is a simple introduction to the most important network concepts for ethical hacking. This is a huge subject and is recommended to learn from different sources like courses, books and certifications like [Cisco CCNA](#) or [CompTIA Network+](#). Also there is a ton of **free training** out there, I recommend to [check this list](#) later.

Objectives

- Understand network basic concepts

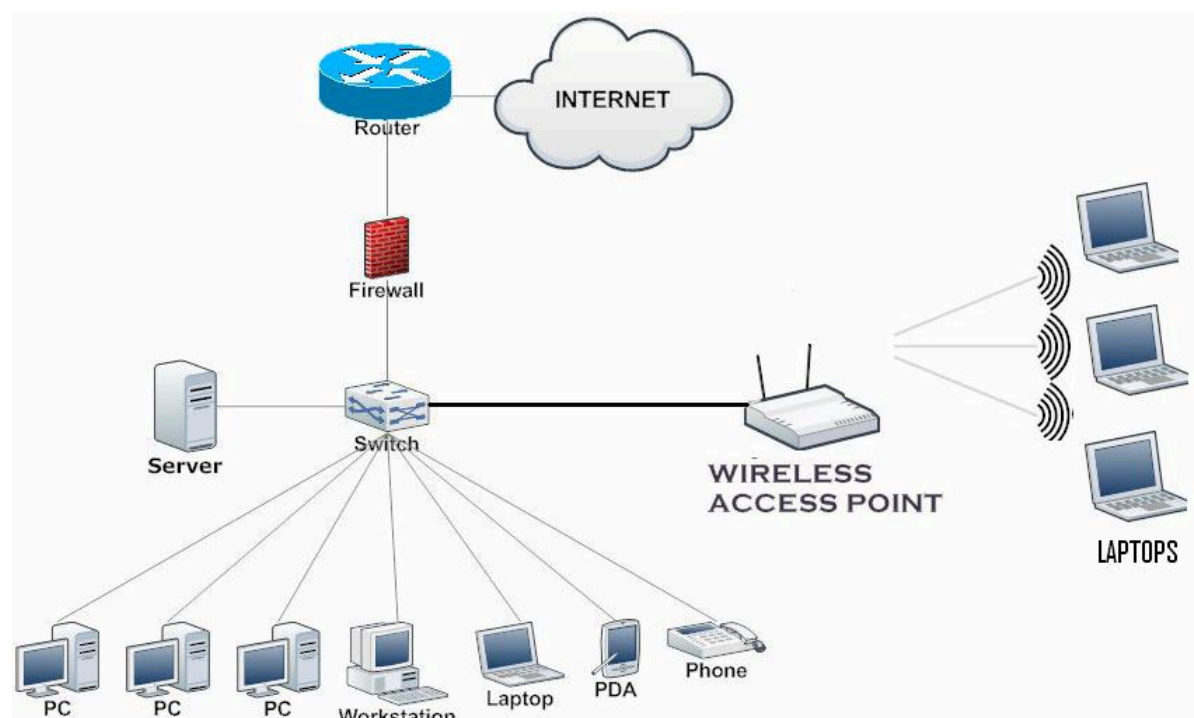
This module follows the order:

1. Introduction
2. IP and MAC Addresses
3. Subnetting
4. TCP, UDP and 3-Way-Handshake
5. Ports & Protocols
6. OSI Model

1. Introduction

So, what the heck is a Network?

A network consists of two or more computers that are linked in order to share resources. Computer networks are the basis of communication in IT. They are used in a huge variety of ways and can include many different types of network. A computer network is a set of computers that are connected together so that they can share information. The earliest examples of computer networks are from the 1960s, but they have come a long way in the half-century since then.



LAN Network Topology - SOHO / Small Home Network

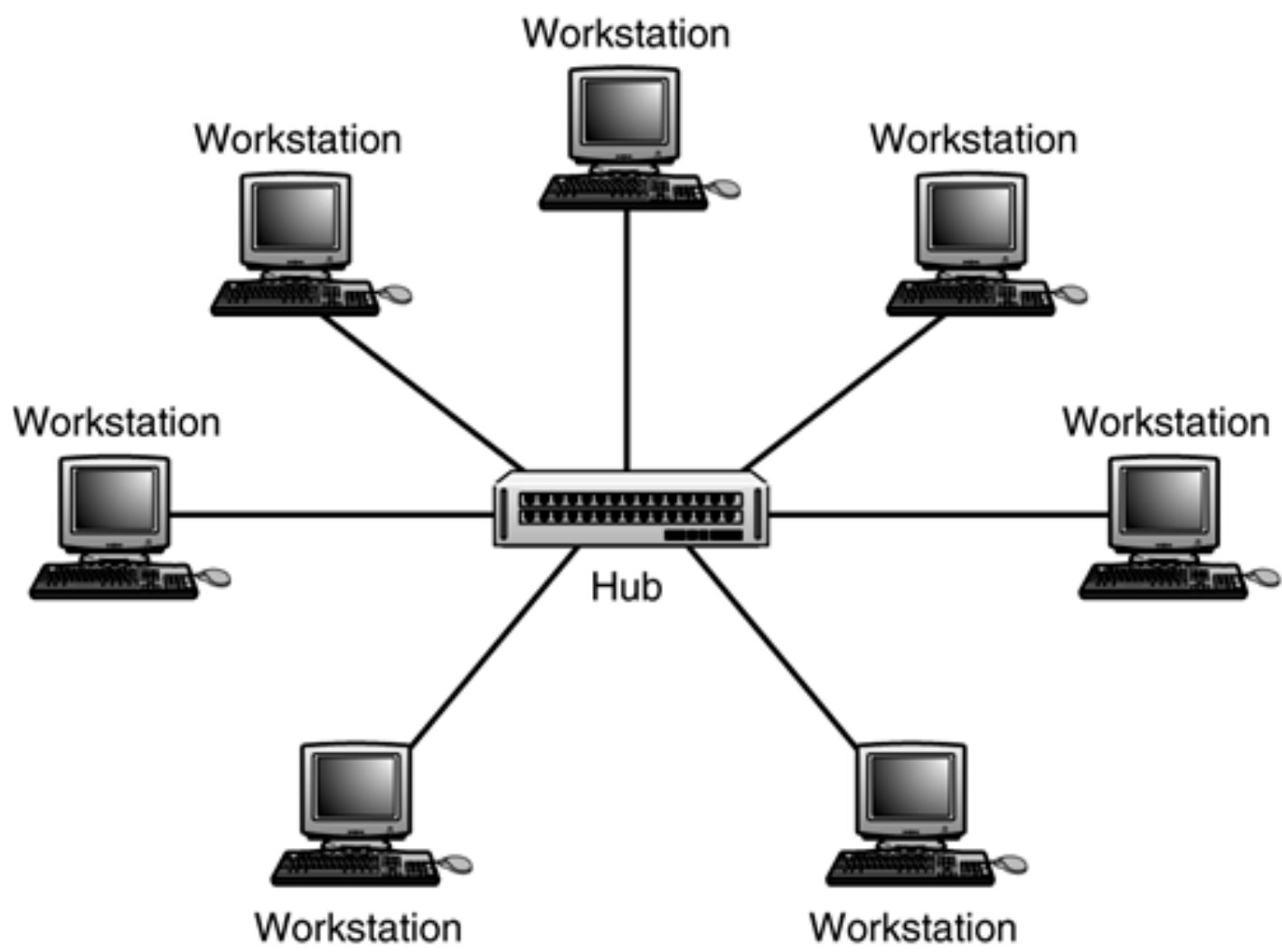
Two very common types of networks include: LAN (Local Area Network) and WAN (Wide Area Network)

Topologies

There are many different types of network, which can be used for different purposes and by different types of people and organization. Here are some of the network types that you might come across:

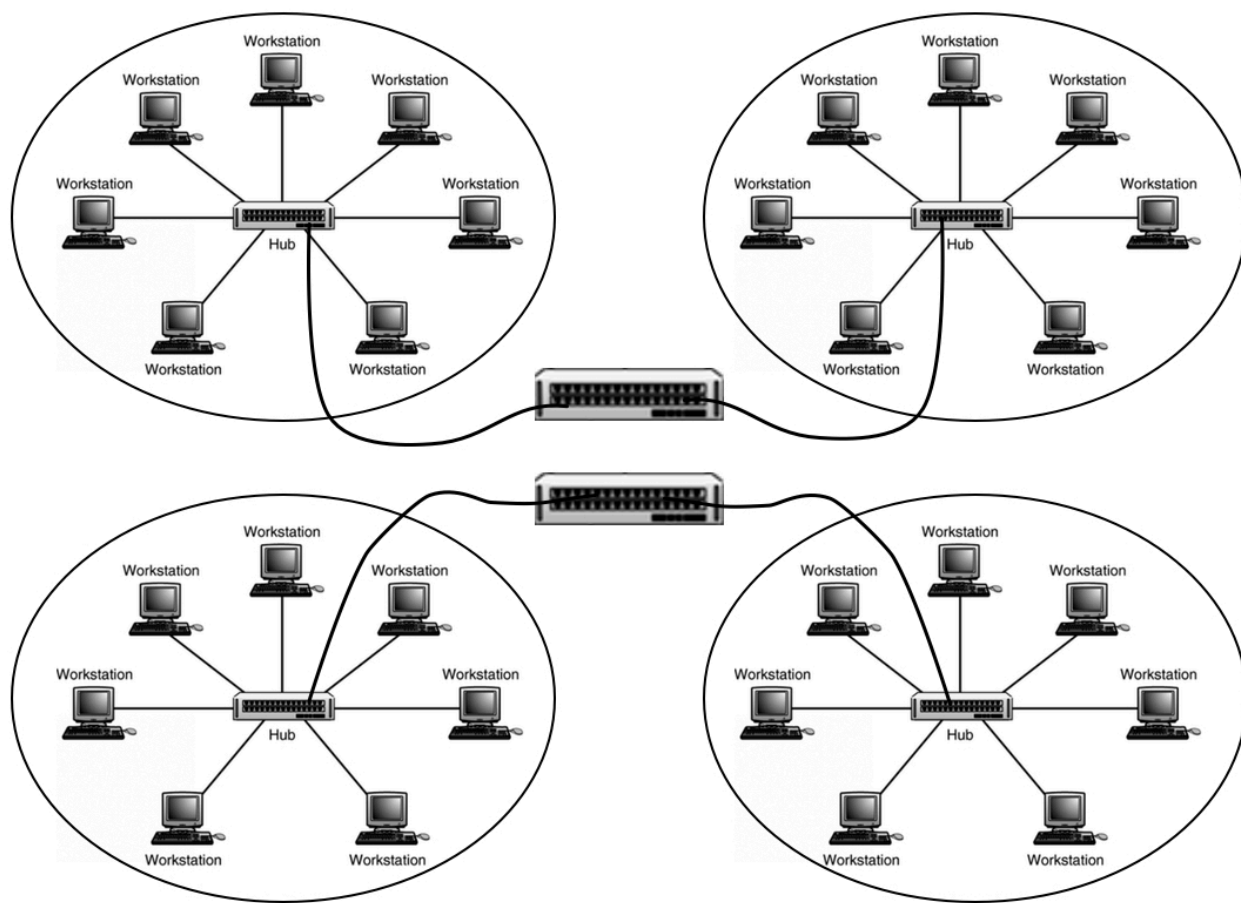
LAN - Local Area Network

- A LAN is a network that has a logical and physical borders that a computer can broadcast

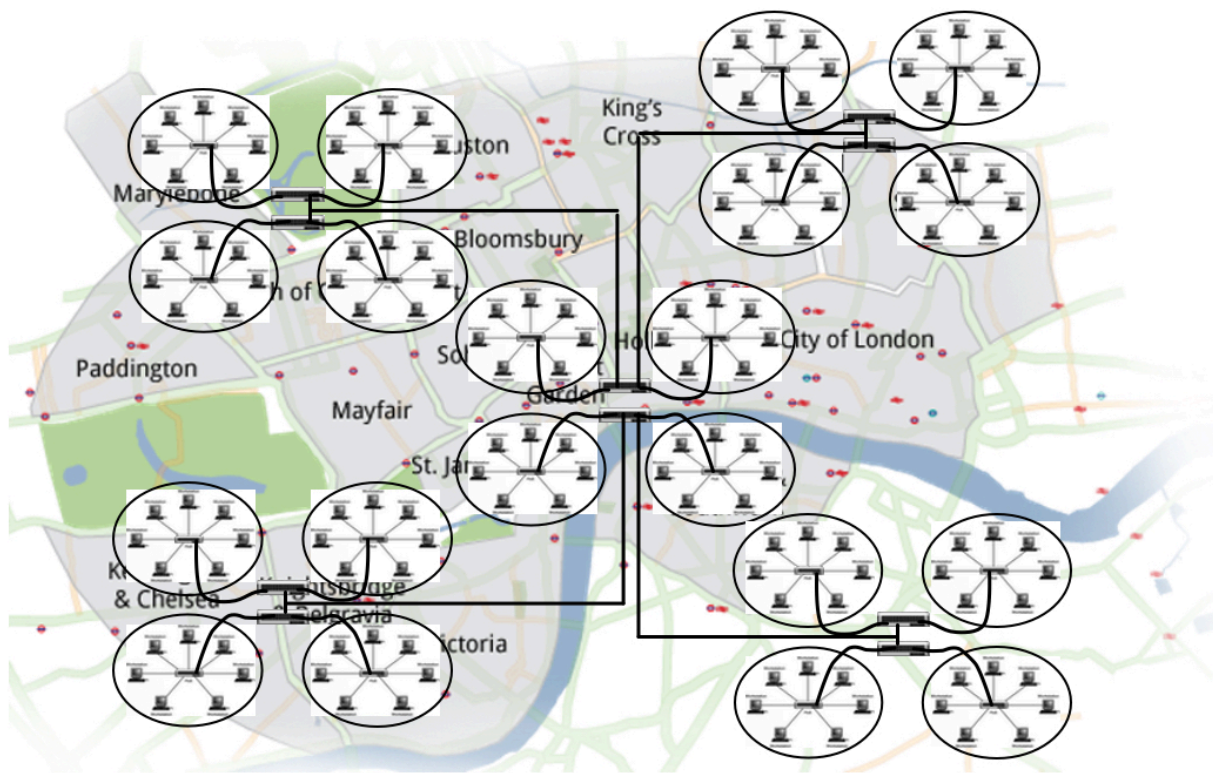


WAN - Wide Area Network

- WAN is a multiple LANs or additional WANs with routing functionality for interconnectivity.



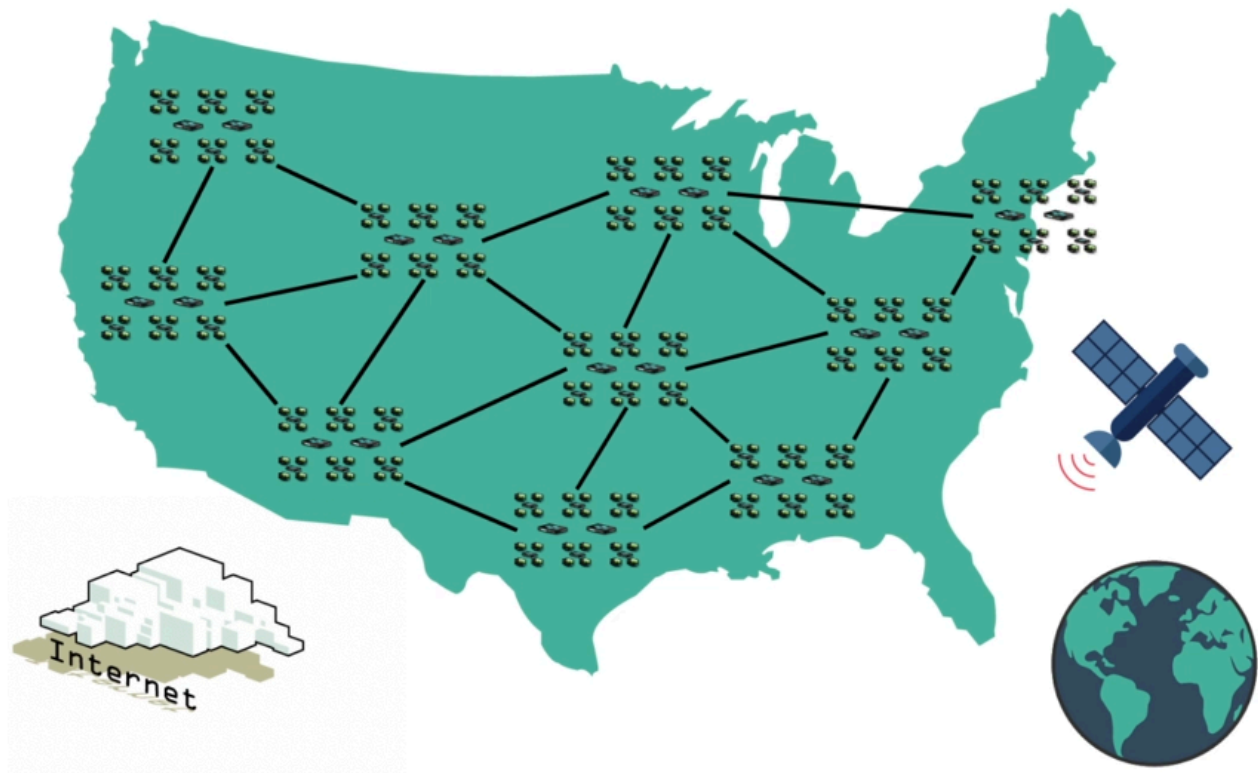
MAN - Metropolitan Area Network



Internet

Connecting WANs through WANs until complete the entire world = Internet.

- The protocol which runs the internet is TCP/IP
- As long you're using legitimate IPv4 address or IPv6

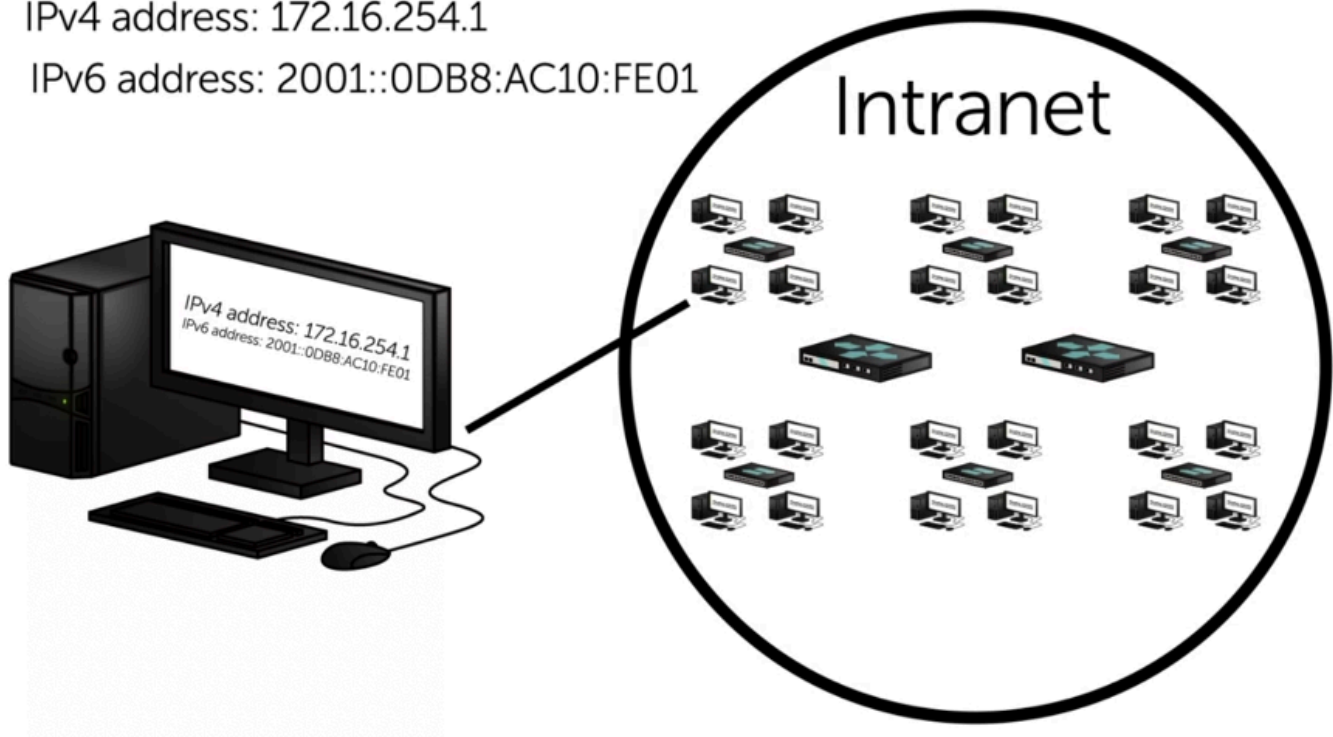


Intranet

If you're using the TCP/IP stack and making your own LAN or WAN = Intranet.

- Intranet is a private network which still runs TCP/IP

IPv4 address: 172.16.254.1
 IPv6 address: 2001::0DB8:AC10:FE01

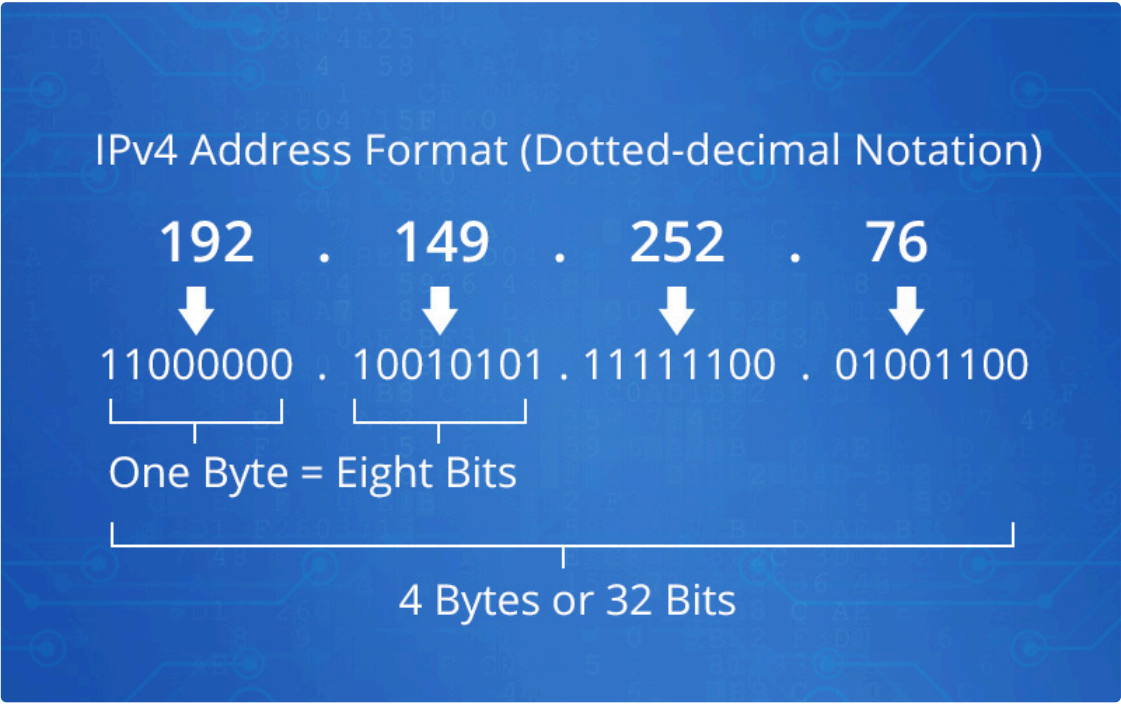


Common Terms in Networking

- **IP (internet protocol) address:** the network address of the system across the network, which is also known as the Logical Address).
- **MAC address:** the MAC address or physical address uniquely identifies each host. It is associated with the Network Interface Card (NIC).
- **Open system:** an open system is connected to the network and prepared for communication.
- **Closed system:** a closed system is not connected to the network and so can't be communicated with.
- **Port:** a port is a channel through which data is sent and received.
- **Nodes:** nodes is a term used to refer to any computing devices such as computers that send and receive network packets across the network.
- **Network packets:** the data that is sent to and from the nodes in a network.
- **Routers:** routers are pieces of hardware that manage router packets. They determine which node the information came from and where to send it to. A router has a routing protocol which defines how it communicates with other routers.
- **Network address translation (NAT):** a technique that routers use to provide internet service to more devices using fewer public IPs. A router has a public IP address but devices connected to it are assigned private IPs that others outside of the network can't see.
- **Dynamic host configuration protocol (DHCP):** assigns dynamic IP addresses to hosts and is maintained by the internet service provider.
- **Internet service providers (ISP):** companies that provide everyone with their internet connection, both to individuals and to businesses and other organizations.

2. IP & MAC Address

What is an IP Address (Internet Protocol)?



An IP address is a unique address that identifies a device on the internet or a local network. IP stands for "Internet Protocol," which is the set of rules governing the format of data sent via the internet or local network.

Check your local IP address

1. If you are using Linux or MacOS you can open your terminal and type `ifconfig` command
2. For Windows machine you can open up the cmd prompt or powershell, then type `ipconfig /all`

```
ubuntu@primary:~$ ifconfig
enp0s2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.64.3 netmask 255.255.255.0 broadcast 192.168.64.255
    inet6 fe80::c83b:ccff:fe0e:1069 prefixlen 64 scopeid 0x20<link>
    ether ca:3b:cc:0e:10:69 txqueuelen 1000 (Ethernet)
    RX packets 204 bytes 45443 (45.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 175 bytes 30287 (30.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 88 bytes 6696 (6.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 88 bytes 6696 (6.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

- **inet IPv4:** 192.168.64.3
 - **inet** --> The inet (Internet protocol family) show the local IP address. This is IP version 4 (IPv4) Using 32-bit decimal number.
- **inet6 IPv6:** fe80::c83b:ccff:fe0e:1069
 - **inet6** --> Is a new version of IP (IPv6), using 128 bits hexadecimal value.
- **ether** --> MAC address - unique identifier assigned to a network interface controller (NIC)

More about the IPv4 decimal value:

IPv4 = 32 bits range (4 octets of 8 bits, from 0-255 each(4))

11000000	.10101000	.01000000	.00000011	[IPv4 binary]
192	.168	.64	.3	[IPv4 decimal]

The arithmetic behind IPv4:

- One octet have 8 bits:

0 or 1	0 or 1	0 or 1	0 or 1	0 or 1	0 or 1	0 or 1	0 or 1
8th bit	7th bit	6th bit	5th bit	4th bit	3rd bit	2nd bit	1st bit
128 (2^7)	64 (2^6)	32 (2^5)	16 (2^4)	8 (2^3)	4 (2^2)	2 (2^1)	1 (2^0)

Here is how binary octets convert to decimal: The right most bit, or least significant bit, of an octet holds a value of 2^0. The bit just to the left of that holds a value of 2^1. This continues until the left-most bit, or most significant bit, which holds a value of 2^7. So if all binary bits are a one, the decimal equivalent would be 255 as shown here:

11111111

| | | | | | | |

(128 +64 +32 +16 +8 +4 +2 +1) --> 255

Example of octet conversion:
IP Address: 192.168.64.3

To calculate the first octet (192.), from binary format to decimal:

1286432168421

| | | | | | | |

11000000

| | | | | | | |

128+ 64+ 0+ 0+ 0+ 0+ 0+ 0 = 192 ---> final value (firt octet IPv4 in decimal)

- Take the IP: 192.168.64.3
- The first octet 192 in 8-bit binary is 11000000.
- Only the 8th and 7th bit is on and the rest of them (6th to 1st bit) is off, meaning the decimal value is the final sum of these values: 128 + 64 = 192

⚠ Why? Computers see everything in terms of binary!!; on and off.

IPv4 and IPv6

IPv4

Deployed 1981

32-bit IP address

4.3 billion addresses

Addresses must be reused and masked

Numeric dot-decimal notation

192.168.5.18

DHCP or manual configuration

IPv6

Deployed 1998

128-bit IP address

7.9x10²⁸ addresses

Every device can have a unique address

Alphanumeric hexadecimal notation

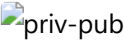
50b2:6400:0000:0000:6c3a:b17d:0000:10a9

(Simplified - 50b2:6400::6c3a:b17d:0:10a9)

Supports autoconfiguration

Private and Public IP Addresses

All IPv4 addresses can be divided into two major groups: **global (or public, external)** - this group can also be called 'WAN addresses' — those that are used on the Internet, and **private (or local, internal) addresses** — those that are used in the local network (LAN).



More about Private IP addresses:

Private (internal) addresses are not routed on the Internet and no traffic can be sent to them from the Internet, they only supposed to work within the local network. Private addresses include IP addresses from the following subnets:

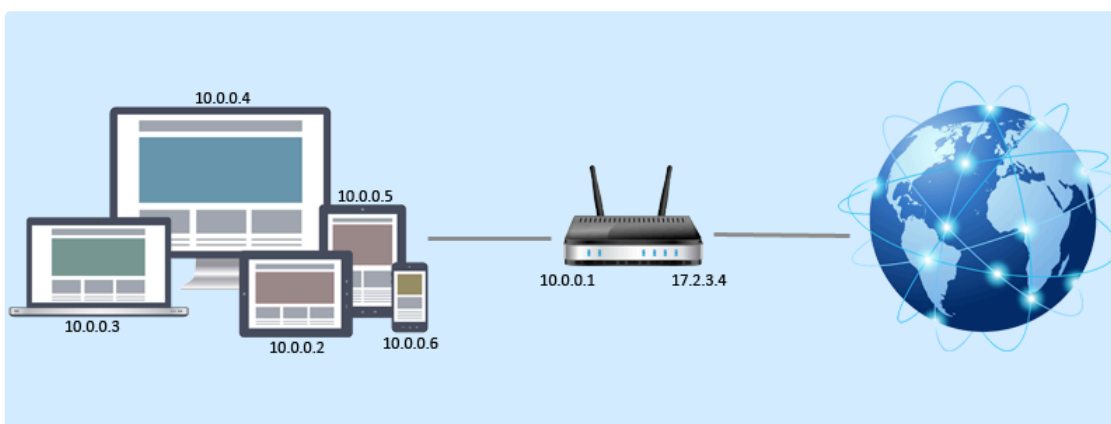
PRIVATE IP ADDRESS

(are not used anywhere on public internet, reserved for private LANs)

Network Class	Network Numbers	Network Mask	No. of Networks	No. of Hosts per Network
CLASS A	10.0.0.0	255.0.0.0	126	16,646,144
CLASS B	172.16.0.0 to 172.31.0.0	255.255.0.0	16,383	65,024
CLASS C	192.168.0.0 to 192.168.255.255	255.255.255.0	2,097,151	254
LOOPBACK (localhost)	127.0.0.0 to 127.0.0.7	255.255.255.0	-	-

NAT - Network Address Translation

NAT stands for network address translation. It's a way to map multiple local private addresses to a public one before transferring the information. Organizations that want multiple devices to employ a single IP address use NAT, as do most home routers.



1. Static NAT

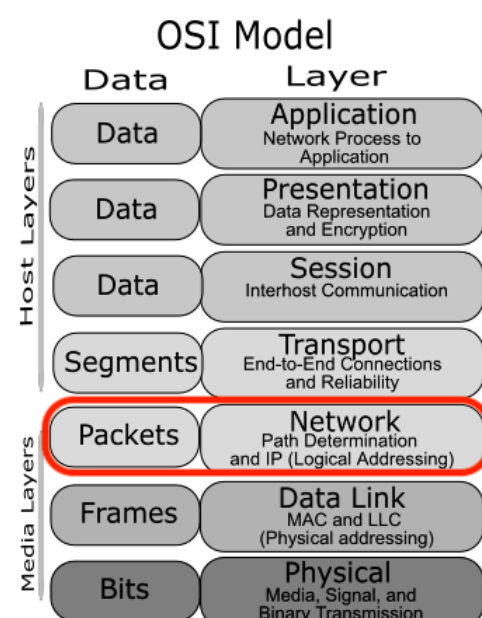
When the local address is converted to a public one, this NAT chooses the same one. This means there will be a consistent public IP address associated with that router or NAT device.

2. Dynamic NAT

Instead of choosing the same IP address every time, this NAT goes through a pool of public IP addresses. This results in the router or NAT device getting a different address each time the router translates the local address to a public address.

⚠ IP Addresses operates on Layer 3 of OSI Model

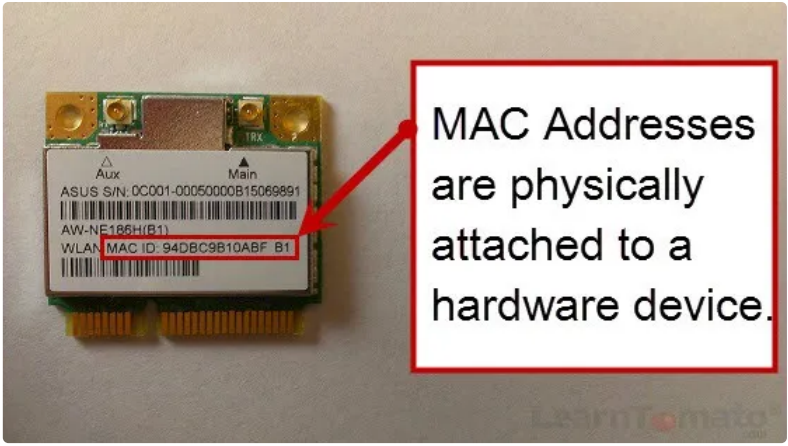
Note: This module will cover OSI model later.



3. Subnetting

Why subnetting?

The way IP addresses are constructed makes it relatively simple for Internet routers to find the right network to route data into. However, in a Class A network (for instance), there could be millions of connected devices, and it could take some time for the data to find the right device. This is why subnetting comes in handy: subnetting narrows down the IP address to usage within a range of devices.



The first three bytes of a MAC address were originally known as OUI's, or Organizational Unique Identifiers. Each manufacturer of networking equipment was assigned an OUI, and was free to assign their own numbers in that block.

OUI	NIC

00:0c:29:99:98:ca	

Checking vendor behind MAC addresse

1. Check your MAC address use the command `ifconfig` (Linux) or `/ipconfig` (Windows)□

```
purple@purple-vm:~$ ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.1.14  netmask 255.255.255.0  broadcast 192.168.1.255
    inet6 fe80::113f:2f5e:4595:94b7  prefixlen 64  scopeid 0x20<link>
    ether 00:0c:29:99:98:ca  txqueuelen 1000  (Ethernet)
    RX packets 159496  bytes 240175618 (240.1 MB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 17535  bytes 1282034 (1.2 MB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 296  bytes 26617 (26.6 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 296  bytes 26617 (26.6 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0
```

2. Copy and save the **first three bytes** of your address. (*The first three bytes from image above is 00:0c:29*)
3. Validate the information by performing a **MAC Address Lookup** on the internet. For this example I'm using: <https://aruljohn.com/>

ENTER MAC ADDRESS OR OUI (FIRST 6 DIGITS)

SELECT LOOKUP TYPE: ☒ LOOKUP MAC ☐ LOOKUP VENDOR

example: 00:0B:14

This database was last updated on 10 April 2021

Results for MAC address 000C29

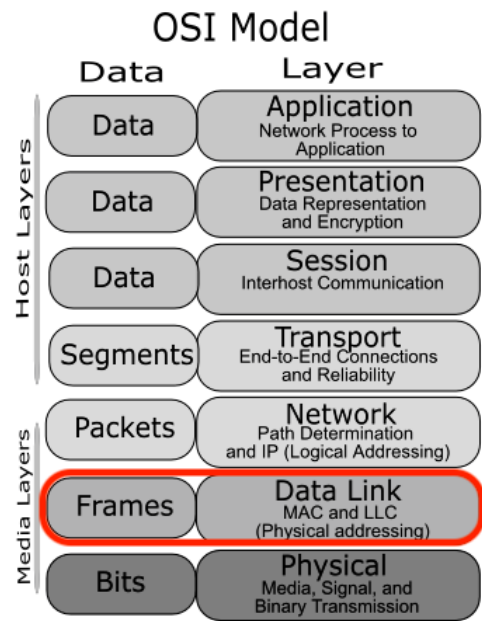
Found 1 results.

MAC Address/OUI	Vendor {Company}
000C29	VMware, Inc.

4. As you can see the OUI lookup identify a virtual network interface provided by VMware

So, to summarize, the **first three bytes** are assigned to a manufacturer of networking equipment and the manufacturer assigns the last three bytes of an address.

⚠️ MAC Addresses operates on Layer 2 of OSI Model



4. TCP/IP, UDP and 3-Way-Handshake

Transmission Control Protocol/Internet Protocol (TCP/IP)

- What is TCP used for?

TCP enables data to be transferred between applications and devices on a network. It is designed to break down a message, such as an email, into packets of data to ensure the message reaches its destination successfully and as quickly as possible.

- What does TCP mean?

TCP means Transmission Control Protocol, which is a communications standard for delivering data and messages through networks. TCP is a basic standard that defines the rules of the internet and is a common protocol used to deliver data in digital network communications.

- The TCP/IP model consists of several types of protocols, including:
 - TCP and IP
 - Address Resolution Protocol (ARP)
 - Internet Control Message Protocol (ICMP)
 - Reverse Address Resolution Protocol (RARP)
 - User Datagram Protocol (UDP)

TCP/IP Layers	TCP/IP Protocols				
Application Layer	HTTP	FTP	Telnet	SMTP	DNS
Transport Layer	TCP		UDP		
Network Layer	IP	ARP	ICMP	IGMP	
Network Interface Layer	Ethernet	Token Ring		Other Link-Layer Protocols	

TCP/IP Model

TCP is the most commonly used of these protocols and accounts for the most traffic used on a TCP/IP network. **UDP is an alternative to TCP that does not provide error correction, is less reliable, and has less overhead, which makes it ideal for streaming.**

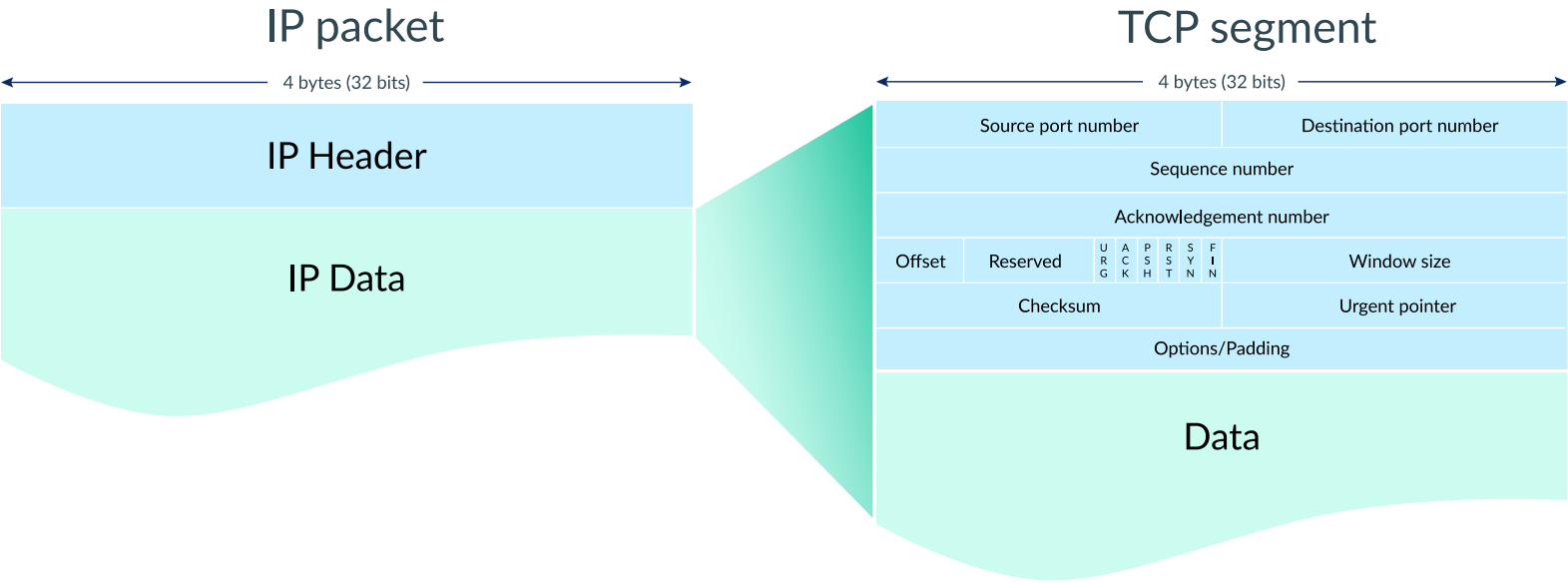
The User Datagram Protocol (UDP)

Is a lightweight data transport protocol that works on top of IP. UDP provides a mechanism to detect corrupt data in packets, but it does not attempt to solve other problems that arise with packets, such as lost or out of order packets. That's why UDP is sometimes known as the Unreliable Data Protocol. UDP is simple but fast, at least in comparison to other protocols that work over IP. It's often used for time-sensitive applications (such as real-time video streaming) where speed is more important than accuracy.

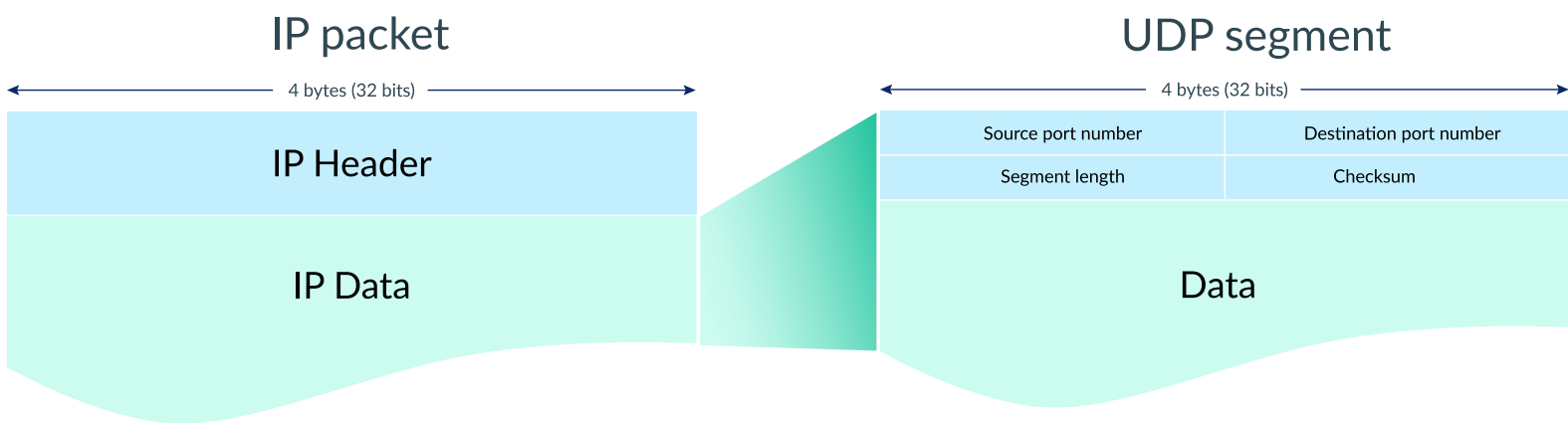
- On Linux and Unix systems you can issue the `lsof` command to see which processes is using UDP ports

```
$ sudo lsof -i -n -P | grep UDP
launchd          1  IPv4  UDP *:137
launchd          1  IPv4  UDP *:138
syslogd         45  IPv4  UDP *:54465
mDNSResponder  186  IPv4  UDP *:5353
mDNSResponder  186  IPv6  UDP *:5353
mDNSResponder  186  IPv4  UDP *:65327
mDNSResponder  186  IPv6  UDP *:65327
mDNSResponder  186  IPv4  UDP *:55657
mDNSResponder  186  IPv6  UDP *:55657
Google        12306  IPv6  UDP *:5353
```

The TCP format



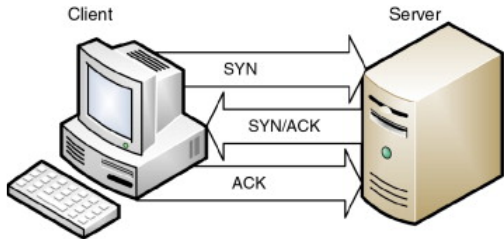
The UDP format



TCP Handshake

TCP uses a three-way handshake to establish a reliable connection. The connection is full duplex, and both sides synchronize (SYN) and acknowledge (ACK) each other. The exchange of these four flags is performed in three steps:

1. SYN
2. SYN-ACK
3. ACK



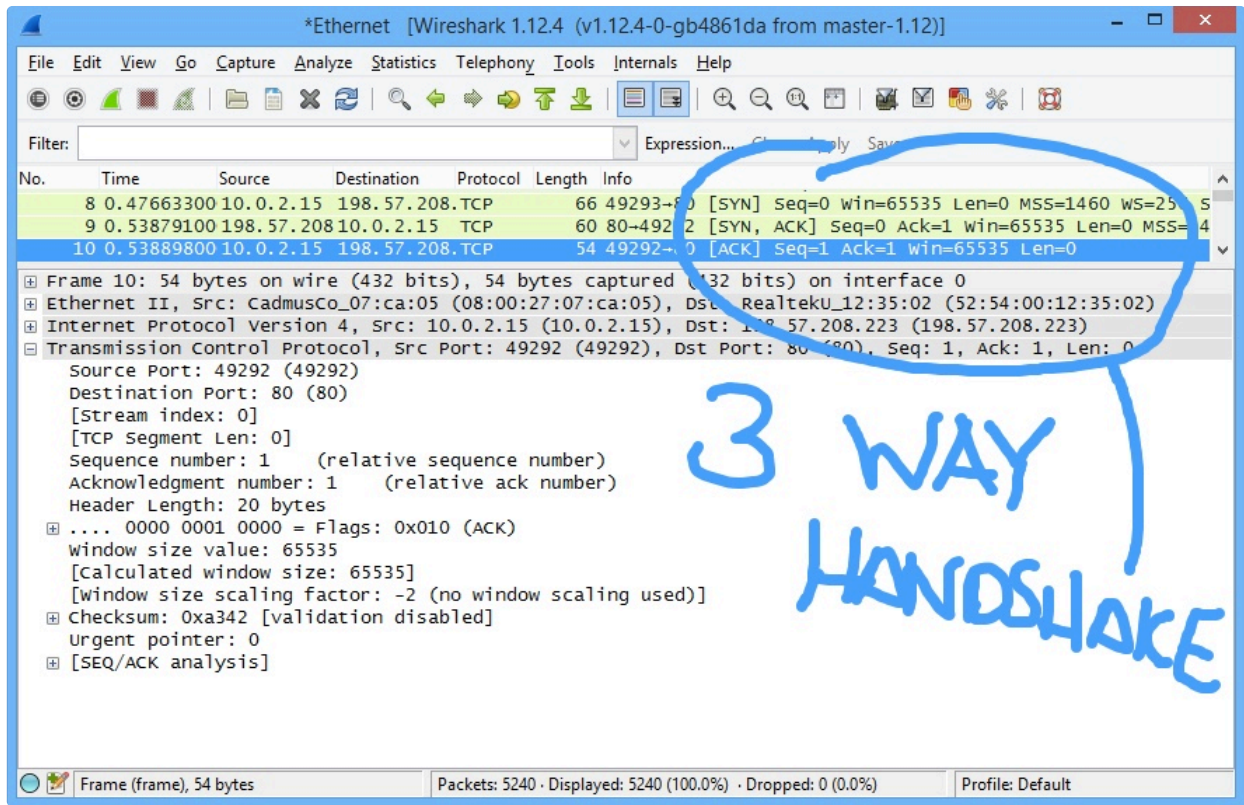
The three message mechanism is designed so that two computers that want to pass information back and forth to each other can negotiate the parameters of the connection before transmitting data such as HTTP browser requests.

More TCP Flags

Flag	Name	Function
SYN	Synchronize	Set during initial communication. Negotiating of parameters and sequence numbers
ACK	Acknowledgment	Set as an acknowledgement to the SYN flag. Always set after initial SYN
RST	Reset	Forces the termination of a connection (in both directions)
FIN	Finish	Ordered close to communications
PSH	Push	Forces the delivery of data without concern for buffering
URG	Urgent	Data inside is being sent out of band. Example is cancelling a message

Capturing 3 Way handshakes (Example)

- The figure below shows the 3-way-handshake packets captured by [Wireshark](#)



5. Ports & Protocols

What is a Port?

In computer networking, a port is a communication endpoint. At the software level, within an operating system, a port is a logical construct that identifies a specific process or a type of network service.

The most common ports

As a penetration tester or ethical hacker you should be familiar with the common ports and protocols used by popular services.

Port Numbers

- Internet Assigned Numbers Authority (IANA) - maintains Service Name and Transport Protocol Port Number Registry which lists all port number reservations

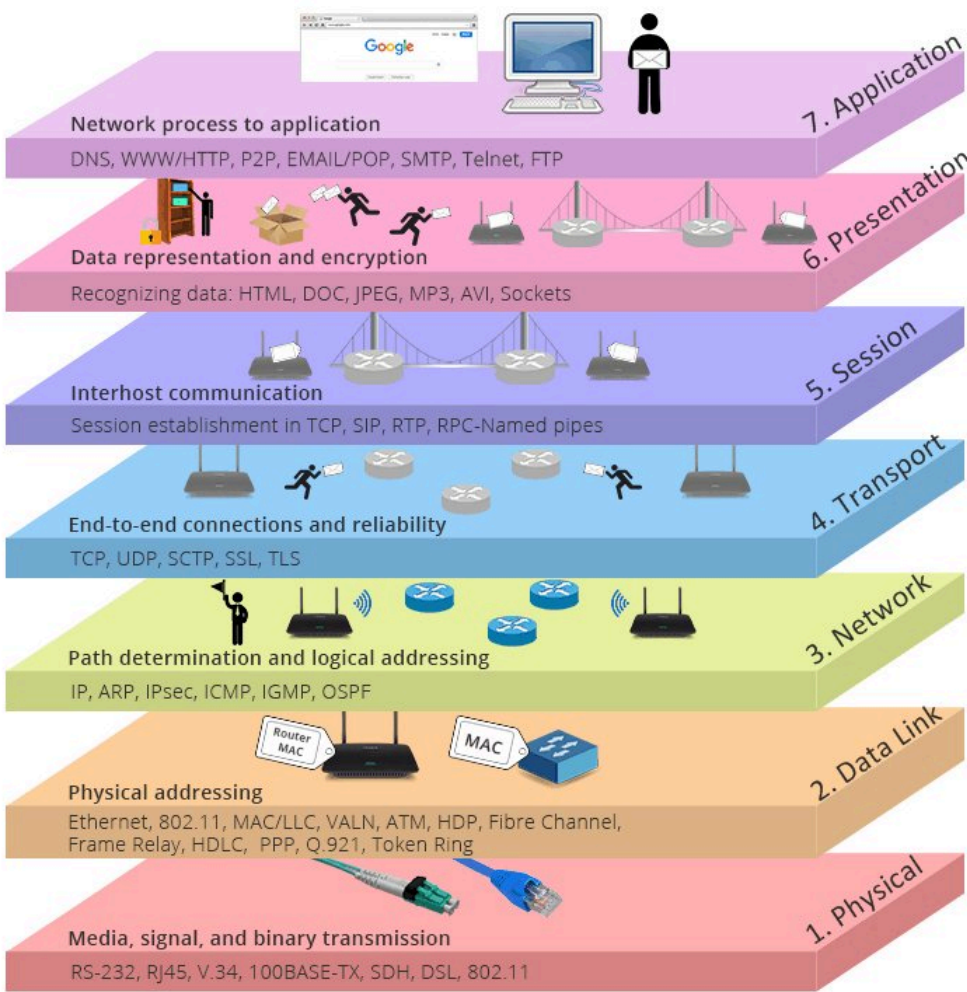
- Ranges
 - **Well-known ports** - 0 - 1023
 - **Registered ports** - 1024 - 49,151
 - **Dynamic ports** - 49,152 - 65,535

Port Number	Protocol	Transport Protocol
20/21	FTP	TCP
22	SSH	TCP
23	Telnet	TCP
25	SMTP	TCP
53	DNS	TCP/UDP
67	DHCP	UDP
69	TFTP	UDP
80	HTTP	TCP
110	POP3	TCP
135	RPC	TCP
137-139	NetBIOS	TCP/UDP
143	IMAP	TCP
161/162	SNMP	UDP
389	LDAP	TCP/UDP
443	HTTPS	TCP
445	SMB	TCP
514	SYSLOG	UDP

- A service is said to be **listening** for a port when it has that specific port open
- Once a service has made a connection, the port is in an **established** state
- `netstat` command:
 - Shows open ports on computer
 - `netstat -an` displays connections in numerical form
 - `netstat -b` displays executables tied to the open port (admin only)

6. OSI Model

OSI Model is a hypothetical networking framework that uses specific protocols and mechanisms in every layer of it. This model is used to divide the network architecture into seven different layers conceptually. These layers are:

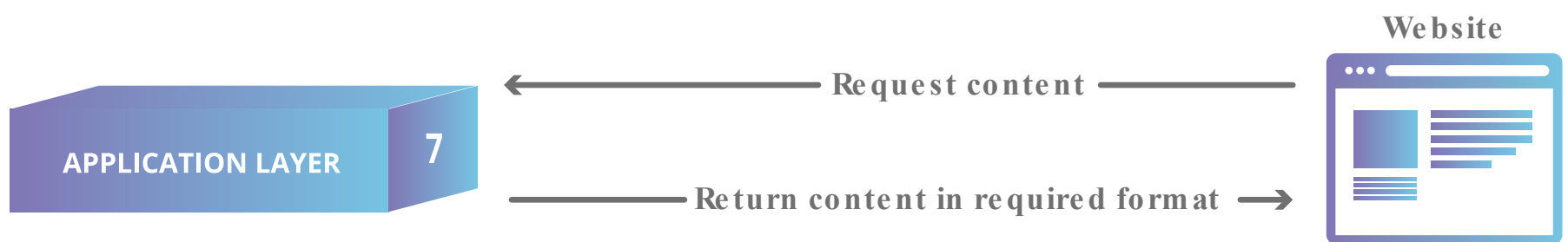


There also involves some security postures and mechanisms that a security professional must know to detect and put the security method effectively in every layer.

More about the Layers:

Layer 7 - Application

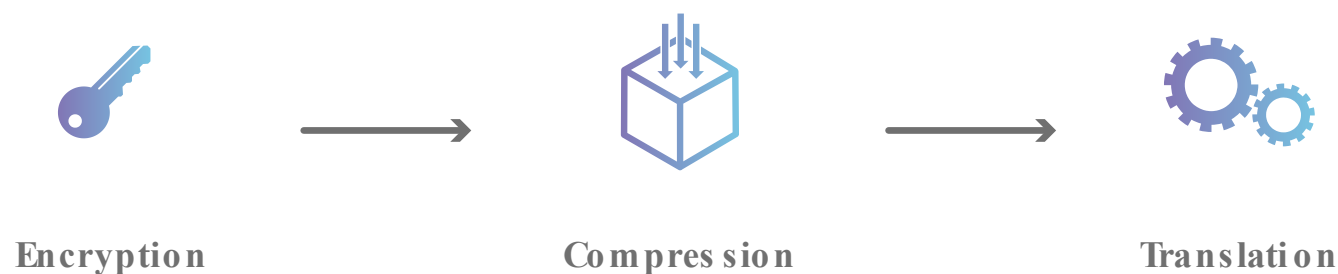
Application Layer



- This is the only layer that directly interacts with data from the user. Software applications like web browsers and email clients rely on the application layer to initiate communications. But it should be made clear that client software applications are not part of the application layer; rather the application layer is responsible for the protocols and data manipulation that the software relies on to present meaningful data to the user. Application layer protocols include HTTP as well as SMTP (Simple Mail Transfer Protocol is one of the protocols that enables email communications).

Layer 6 - Presentation

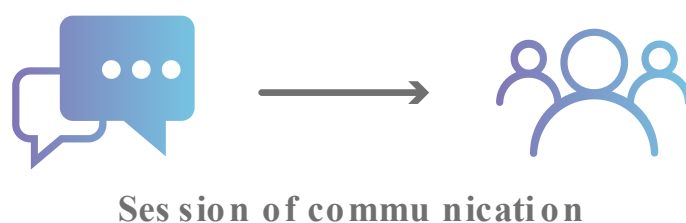
The Presentation Layer



- This layer is primarily responsible for preparing data so that it can be used by the application layer; in other words, layer 6 makes the data presentable for applications to consume. The presentation layer is responsible for translation, encryption, and compression of data.

Layer 5 - Session Layer

The Session Layer



- This is the layer responsible for opening and closing communication between the two devices. The time between when the communication is opened and closed is known as the session. The session layer ensures that the session stays open long enough to transfer all the data being exchanged, and then promptly closes the session in order to avoid wasting resources.

Layer 4 - Transport Layer

Transport Layer



- Layer 4 is responsible for end-to-end communication between the two devices. This includes taking data from the session layer and breaking it up into chunks called segments before sending it to layer 3. The transport layer on the receiving device is responsible for reassembling the segments into data the session layer can consume.
- The transport layer is also responsible for flow control and error control. Flow control determines an optimal speed of transmission to ensure that a sender with a fast connection doesn't overwhelm a receiver with a slow connection. The transport layer performs error control on the receiving end by ensuring that the data received is complete, and requesting a retransmission if it isn't.

Layer 3 - Network Layer

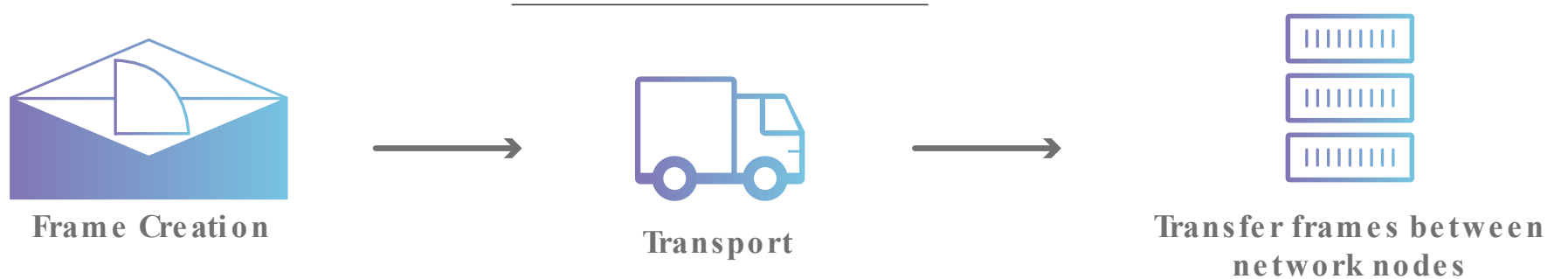
The Network Layer



- The network layer is responsible for facilitating data transfer between two different networks. If the two devices communicating are on the same network, then the network layer is unnecessary. The network layer breaks up segments from the transport layer into smaller units, called packets, on the sender's device, and reassembling these packets on the receiving device. The network layer also finds the best physical path for the data to reach its destination; this is known as routing.

Layer 2 - Data Link Layer

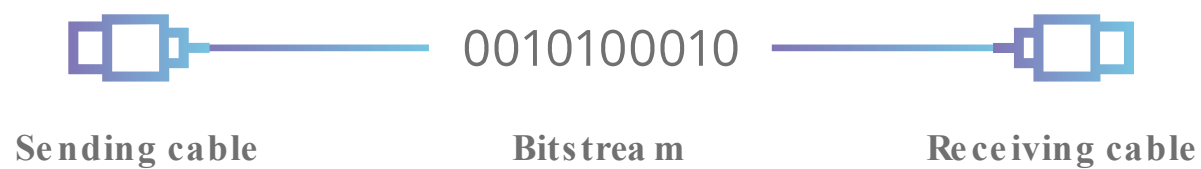
The Data Link Layer



- The data link layer is very similar to the network layer, except the data link layer facilitates data transfer between two devices on the SAME network. The data link layer takes packets from the network layer and breaks them into smaller pieces called frames. Like the network layer, the data link layer is also responsible for flow control and error control in intra-network communication (The transport layer only does flow control and error control for inter-network communications).

Layer 1 - Physical Layer

The Physical Layer



- This layer includes the physical equipment involved in the data transfer, such as the cables and switches. This is also the layer where the data gets converted into a bit stream, which is a string of 1s and 0s. The physical layer of both devices must also agree on a signal convention so that the 1s can be distinguished from the 0s on both devices.